Carson Sila Wambua

Junior Software Developer | Java, Python, Spring Boot, React/Next.js | REST APIs, Docker, CI/CD, LLM and RAG Applications

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PROFESSIONAL SUMMARY

Junior software developer who builds and ships tested, documented services end to end. Built Java 21 and Spring Boot microservices for a fintech distributor-financing platform (payment callbacks, ledger reconciliation, approval workflows, JWT security), and Python applications that use LLM APIs and retrieval (RAG) for real decision-support use cases. Comfortable with REST APIs, relational databases, Docker and CI/CD, and used to working with QA, DevOps, business analysts and non-technical stakeholders. BSc Information Technology completing November 2026; available immediately and open to relocation.

TECHNICAL SKILLS

- **Backend:** Java 21, Spring Boot 3.4, Spring Cloud (Eureka, Config Server, Gateway), Spring Security/JWT, REST APIs, OpenFeign, OpenAPI/Swagger, microservices and event-driven design (saga choreography)
- **Python:** Pandas, NumPy, Scikit-Learn, Streamlit, Plotly, Pytest
- **AI and LLM:** LLM API integration (Gemini), retrieval-augmented generation (RAG), prompt design, anomaly detection, Random Forest, model evaluation
- **Databases:** PostgreSQL, MySQL, SQL query design, data modelling
- **Frontend:** React/Next.js and Angular (working knowledge), TypeScript/JavaScript
- **Delivery and tooling:** Docker, Maven, Jenkins (CI/CD), Git/GitHub, Linux, Postman/Swagger testing
- **Fintech:** M-Pesa C2B/B2B and core-banking callbacks, idempotent transaction handling, multi-tenant KYC onboarding, maker-checker approvals, waterfall repayment allocation
- **Languages:** English (fluent), Kiswahili (fluent)

PROFESSIONAL EXPERIENCE

Backend Software Developer, E&M Technology House | July 2026 – September 2026
Financial Technology and Enterprise Engineering, Nairobi, Kenya | Distributor Financing Platform

- Built the Repayments microservice: M-Pesa and core-banking payment callback listeners with idempotent transaction handling, batch sweep schedulers and ledger reconciliation routines.
- Implemented a waterfall allocation engine that applies incoming payments across penalties, fees, interest and principal and restores revolving credit headroom.
- Built the Onboarding microservice: multi-tenant KYC workflows, document storage and role-based access control for bank admins, manufacturers and distributors; added maker-checker approval controls.
- Standardised inter-service communication with OpenFeign across the Loans, Notifications, Reports and API Gateway services; set up Spring Cloud Config and Eureka service discovery.
- Secured services with Spring Security and JWT and documented APIs with aggregated Swagger/OpenAPI.
- Contributed to the system design: a 14-section design document covering 18 microservices, plus BRD, FRD and architecture diagrams.
- Worked daily with QA, DevOps, business analysts and frontend developers; shipped with Maven, Docker, Jenkins and Git.

Operational Data Analyst Fellow, Power Learn Project (PLP) / Inuka Enterprise | June 2026 – September 2026
Analytics track (part-time), Nairobi, Kenya

- Built OpsPulse AI (see Projects) and presented its business case to a non-technical finance audience.

Network Operations and Telecommunications Engineering Intern, Quavatel Limited | Apr 2024 – Oct 2024

- Monitored 24/7 network telemetry for tier-1 operators (Safaricom, Airtel, Liquid Telecom) under a 99.8% SLA uptime benchmark.
- Contributed to a 28% faster average emergency restoration time; audited GPS fleet-tracking software for 50+ field vehicles.

SELECTED PROJECTS

ClimaCrop Intelligence: Climate-Smart Agri-Fintech Platform (capstone) | Python, Streamlit, Scikit-Learn, Gemini API, Docker

- Built a role-based application for cooperative members, bank/SACCO officers and researchers, using 10 years of climate data (116 stations, 35 counties) and a 40-crop suitability matrix.
- Combined transparent rule-based crop recommendations with a Random Forest audit model and data-confidence tags; built an underwriting engine for loan sizing and DSCR stress tests.
- Integrated a Gemini-based advisory assistant in Swahili and English with an offline rule-based fallback.

OpsPulse AI: Operations Intelligence and SLA Copilot | Python, Streamlit, Scikit-Learn, Plotly, Pytest

- Built Isolation Forest anomaly detection over 1,800+ monthly incident logs, with RAG retrieval over SRE runbooks and automated 5-Whys post-mortems and remediation checklists.
- Covered by a 15-test Pytest suite with a 100% pass rate.
- Wrote a CFO business case with projected figures (modelled, not measured): MTTR from 45 to 12 minutes, 461% first-year ROI, 53-day payback.

Task Management App (Bedtask) | Spring Boot, Next.js

- Full-stack app: REST backend with task priority enums and Swagger documentation, and a Next.js frontend.

Student Management System | Java, Spring Boot

- REST APIs with exception handling and Spring Security/JWT authentication.

E-Commerce Agricultural Trade Platform (degree capstone)

- Connects smallholder farmers to markets; received top marks from the university evaluation panel.

More projects, live demos and code: portfolio-carsonsila.vercel.app

EDUCATION

Bachelor of Science in Information Technology, Zetech University | 2024 – November 2026 (expected completion)
Software engineering and data analytics focus.

Diploma in Information Technology, Zetech University | 2022 – 2024 | Graduated with Distinction

CERTIFICATIONS

- Microsoft AI and Data Analysis certificate (2026)

LEADERSHIP AND HONORS

- **1st Runner-Up, University Innovation Hackathon, 2025:** co-led the architecture of a campus fintech solution.
- **Youth Ministry Teacher and Programme Leader, Thika (2021 – 2025):** taught and mentored young people and coordinated weekly programmes for 500+ attendees.

References available on request.